

TABLE III
Earth Ground Testers from Different Manufacturers

Specifications		Fluke Fluke 1621	Metrix+ Model: DET 1503	IDEAL 61-920
General specifications	Measuring functions	Three-pole earth ground resistance, two-pole AC resistance of a conductor, interference voltage	0.00-2000Ω	Ground resistance, ground leakage current
	Measuring rate	Two measurements/second	5sec/V 2sec	0/120/1200Ω
	Safety	IEC/EN 61010-1, 600V CAT II, pollution degree 2	IEC 61010-2-031	IEC61010-1: 2001 (CAT IV 300V Pollution degree 2), IEC61010-2-032: 2002
	IP rating	IP 40; IEC/EN 60529	IEC60529 (Ip54)	IP 40
	Dimensions	113×54×216mm including holster	170×120×79mm	246×119×53mm
	Weight (gm)	850	675	771
Electrical specifications	Test voltage	3.7kV	0-200V	—
	Measuring time	8 seconds (average from when start is pressed)	10 sec-10 min	—
	Electromagnetic compatibility	Emission: IEC/EN 61326 Class B, Immunity: IEC/EN 61326 Annex C	Yes	—
	Interference voltage display	V _{max} 30Veff Common mode rejection >80dB at 50Hz and 60Hz Ri 680kΩ Measuring uncertainty <10% for pure AC and DC signals	Yes	Yes
Measuring range resolution		Resolution 0.01Ω, display range 0-19.99Ω	0.01Ω	
		Resolution 0.1Ω, display range 20-199.9Ω	0.1Ω	
		Resolution 1Ω, display range 200-1999Ω	1Ω	
RE resistance measurement	Open circuit voltage (Vac)	23V to 24V AC	220V	—
	Short circuit current (mA)	>50mA AC	1.3mA	0.1mA
	Measuring frequency (Hz)	128Hz	50/60Hz	2400Hz
	Permissible overload (Veff)	250Veff	300V AC 30 seconds	—

frequency offset, which may be measured via a *beat* note of the sent-out and received beam.

Fig. 4 shows a distance meter.

Vibration tester

Machines are used in nearly every aspect of our daily life; from the vacuum cleaner and washing machine we use at home, to the industrial machinery used to manufacture nearly every product we use on a daily basis. When a machine fails or breaks down, the consequences can range from financial loss, personal injury and possible loss of life. For this reason, early detection, identification and correction of problems is

paramount to anyone involved in the maintenance of industrial machinery to ensure continued, safe and productive operation.

Vibration levels give a very good idea of the state of a machine. It is evident from the fact that we naturally touch a rotating machine and feel the vibrations to see if it is running right. Even machines in the best of operating condition will have some vibration because of small, minor defects. Therefore each machine will have a level of vibration that may be regarded as normal or inherent. However, when vibration increases or becomes excessive, some mechanical trouble is usually the reason. Various instruments have

been developed to actually measure a machine's vibration level and assign it a numerical value. Fig. 5 shows a vibration meter and Table IV shows



Fig. 5: Vibration meter